

CS336: Language Modeling from Scratch

Stanford Frontier AI Learning System

Baseline: October 6, 2026. Print companion to the HTML course site:

course map, prerequisites, notation and shapes, glossary,
cheatsheet, crash course.

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Course map

Nineteen scheduled sessions mapped to units without omissions. Baseline: 2026-10-06.

Session skeleton (titles, dates, instructors) comes from SRC-04, a secondary study wiki. It is NOT official evidence. Every session leaf is therefore PLANNED / SOURCE ATTRIBUTION PENDING. Claim class per session: OFFICIAL-SOURCE only where an inspected official artifact backs the leaf, REQUESTED-BRANCH everywhere else.

Block 1 , Foundations (weeks 1-4)

#	Date	Session (re-ported)	Instructor (re-ported)	Units	Claim class
S01	Mar 30	Overview, Tok-enization	Percy Liang	U01	REQUESTED-BRANCH, U01-C03/C04 partial OFFICIAL-SOURCE via SRC-03
S02	Apr 1	PyTorch, Re-source Ac-counting	Percy Liang	U02	REQUESTED-BRANCH
S03	Apr 6	Architectures, Hyperparameters	Tatsu Hashimoto	U03	REQUESTED-BRANCH
S04	Apr 8	Attention Alternatives, MoE	Tatsu Hashimoto	U04	REQUESTED-BRANCH

Block 2 , Systems (weeks 5-8)

#	Date	Session (re-ported)	Instructor (re-ported)	Units	Claim class
S05	Apr 13	GPUs, TPUs	Tatsu Hashimoto	U06	REQUESTED-BRANCH
S06	Apr 15	Kernels, Triton, XLA	Percy Liang	U07	REQUESTED-BRANCH
S07	Apr 20	Parallelism I (data, pipeline)	Percy Liang	U08, U09-C02/C03/C04	REQUESTED-BRANCH
S08	Apr 22	Parallelism II (tensor, expert)	Tatsu Hashimoto	U09	REQUESTED-BRANCH

Block 3 , Scaling (weeks 9-10)

#	Date	Session (re-ported)	Instructor (re-ported)	Units	Claim class
S09	Apr 27	Scaling Laws	Tatsu Hashimoto	U10	REQUESTED-BRANCH (second builder)
S10	Apr 29	Inference	Percy Liang	U11	REQUESTED-BRANCH (second builder)
S11	May 4	Scaling Laws II	Tatsu Hashimoto	U10	REQUESTED-BRANCH (second builder)

Block 4 , Data (weeks 11-12)

#	Date	Session (re-ported)	Instructor (re-ported)	Units	Claim class
S12	May 6	Evaluation	Percy Liang	U12	REQUESTED-BRANCH (second builder)
S13	May 11	Data: Sources, Transformation, Filtering	Percy Liang	U13	REQUESTED-BRANCH (second builder)
S14	May 13	Data: Mixing, Rewriting, SFT	Percy Liang	U14, U15	REQUESTED-BRANCH (second builder)

Block 5 , Alignment (weeks 13-15)

#	Date	Session (re-ported)	Instructor (re-ported)	Units	Claim class
S15	May 18	Alignment: RLHF/DPO	Tatsu Hashimoto	U16	REQUESTED-BRANCH (second builder)
S16	May 20	Alignment: RL Algorithms	Tatsu Hashimoto	U16	REQUESTED-BRANCH (second builder)
S17	May 27	Alignment: RL Systems	Percy Liang	U17	REQUESTED-BRANCH (second builder)

Guest lectures

#	Date	Session (re-ported)	Speaker (re-ported)	Units	Claim class
G1	Jun 1	Guest lecture	Daniel Selsam	U18	REQUESTED-BRANCH (second builder). Content unknown, see G-08. Do not infer from speaker name.
G2	Jun 3	Guest lecture	Dan Fu	U18	REQUESTED-BRANCH (second builder). Content unknown, see G-08. Do not infer from speaker name.

First-half unit/session reconciliation

- S01 maps to U01 (12 concepts, C01-C12). No omissions.
- S02 maps to U02 (12 concepts, C01-C12). No omissions.
- S03 maps to U03 (12 concepts, C01-C12). Hyperparameter optimization

branches that also touch U05-C07 and U05-C10 are cross-linked, not moved.

- S04 maps to U04 (12 concepts, C01-C12). No omissions.
- S05 maps to U06 (12 concepts, C01-C12). No omissions.
- S06 maps to U07 (12 concepts, C01-C12). No omissions.
- S07 maps to U08 (C01-C12) plus U09 pipeline concepts (C02, C03, C04).

The split follows the reported session focus: data parallelism in S07, tensor and expert parallelism in S08. U09 keeps all 12 concepts as one unit for teaching coherence, the map records the session split.

- S08 maps to U09 (C01-C12). No omissions.
- Optimization and training correctness (U05) has no dedicated reported

session in the first half, the course assigns it as a cross-cutting branch anchored in reported Assignment 1 objectives (transformer LM, cross-entropy, training loop, optimizer). It is taught as U05 with all 12 concepts. This placement is a REQUESTED-BRANCH decision, recorded here rather than hidden.

Second-half unit/session reconciliation (2026-10-06, second builder)

- S09 (Scaling Laws) and S11 (Scaling Laws II) map to U10 (C01-C12).

The two-session split follows the reported schedule: fitting and isoFLOP curves in S09, allocation and extrapolation limits in S11.

- S10 (Inference) maps to U11 (C01-C12). No omissions.
- S12 (Evaluation) maps to U12 (C01-C12). No omissions.
- S13 (Data: Sources, Transformation, Filtering) maps to U13 (C01-C12).
- S14 (Data: Mixing, Rewriting, SFT) maps to U14 (C01-C12) plus U15

(C01-C12). The split follows the reported session focus: filtering, dedup, mixing in U14, mid-training and SFT adaptation in U15.

- S15 (Alignment: RLHF/DPO) and S16 (Alignment: RL Algorithms) map to

U16 (C01-C12). The split follows the reported schedule: preference modeling and DPO in S15, PPO/GRPO and reward design in S16.

- S17 (Alignment: RL Systems) maps to U17 (C01-C12): multimodal

interfaces plus the training-system integration the reported session title names.

- G1 and G2 map to U18 (C01-C12). Content unknown per G-08, the unit

teaches the honesty protocol, replication discipline, ablation design, and the oral defense ladder. No content is inferred from speaker names.

- Multimodality has no dedicated reported session, it is taught as the

first half of U17 as a REQUESTED-BRANCH decision, recorded here rather than hidden, with scale-scope limits stated (U17-C12).

Session count check: 17 core + 2 guest = 19. All 19 appear above. Zero sessions unmapped. Zero sessions mapped twice at leaf level except the documented S09/S11, S15/S16, and S14 splits.

Claim class summary (all 19 sessions)

Every session leaf is REQUESTED-BRANCH except U01-C03/C04 (partial OFFICIAL-SOURCE via SRC-03). All leaves are PLANNED / SOURCE ATTRIBUTION PENDING. The second builder verified this on 2026-10-06 against `source_manifest.md` and `source_gaps.md` before writing U10-U18. No inspected official artifact backs any second-half leaf.

Prerequisites

Shared bridge modules P01-P24 live once at `../shared/prerequisites/` and are linked here, never rebuilt. Each unit lesson also carries a local remediation block for the exact objects it uses first.

Unit prerequisite map

Unit	Bridges	What the unit assumes on day one
U01	P01, P02, P06, P13	integer arithmetic, Python loops and dicts, conditional probability, text as sequences
U02	P03, P11, P12, P15	matrix shapes, autodiff graphs, tensor views, FLOPs versus bytes
U03	P05, P11, P12, P14	chain rule, normalization, broadcasting, attention block layout
U04	P04, P14, P15	low-rank structure, attention cost, memory hierarchy
U05	P05, P09, P11, P12	gradients, convexity and smoothness, optimizer state, stable softmax
U06	P12, P15, P16	dtypes and devices, bandwidth and latency, collectives vocabulary
U07	P12, P15	tensor programs, SRAM versus HBM, roofline intuition
U08	P12, P16	autograd semantics, message passing, partitioning
U09	P14, P15, P16	transformer block math, inter-connect topology, consistency

Bridge reading order for this course

P01, P02, P03, P06, P05, P11, P12, P13, P14, P04, P09, P15, P16, P08. (Only the bridges this half uses. P07, P10, P17-P24 serve the second half.)

Diagnostic

Attempt closed-book. Score each item 0 (cannot start), 1 (partial), or 2 (full). Remediation follows the table. The answer key is `keys/diagnostic_key.md`, do not open it first.

D1. Write the UTF-8 bytes of the character "é" (U+00E9) as integers. Link: `../shared/prerequisites/p13_language.md`. D2. A Python dict maps pairs to counts. Write code that returns the pair with the highest count and its count. Link: `../shared/prerequisites/p02_python.md`. D3. For events A and B, write $P(A|B)$ in terms of $P(B|A)$, $P(A)$, $P(B)$. Link: `../shared/prerequisites/p06_probability.md`. D4. X has shape (4, 8) and Y has shape (8, 3). Give the shape of $X @ Y$ and the FLOP count of the product. Link: `../shared/prerequisites/p03_vectors.md`. D5. State the chain rule for $z = f(g(x))$ with one line of working. Link: `../shared/prerequisites/p05_calculus.md`. D6. Draw the forward and backward data flow of $y = xW + b$ and name every saved tensor. Link: `../shared/prerequisites/p11_neural_nets.md`. D7. A tensor has shape (2, 3, 4) and stride (12, 4, 1). How many elements does it hold, and is it contiguous? Link: `../shared/prerequisites/p12_pytorch.md`. D8. Define perplexity of a sequence under a model in one sentence, then give its formula. Link: `../shared/prerequisites/p13_language.md`. D9. Sketch one transformer block and label Q, K, V, the residual path, and the norm placement. Link: `../shared/prerequisites/p14_transformer.md`. D10. A matrix has rank 2. What does that say about its SVD, in one sentence? Link: `../shared/prerequisites/p04_spectral.md`. D11. Write one step of gradient descent for parameters w with learning rate η . Link: `../shared/prerequisites/p09_optimization.md`. D12. A GPU kernel reads 1 GB and does 10 GFLOP. Is it memory-bound or compute-bound on a 1 TB/s, 100 TFLOP/s device? Show the arithmetic. Link: `../shared/prerequisites/p15_hardware.md`. D13. Name the three collectives that move a full gradient to every rank and back in data parallel training. Link: `../shared/prerequisites/p16_distributed.md`. D14. Define cross-entropy $H(p, q)$ for discrete distributions and say what it measures. Link: `../shared/prerequisites/p08_information.md`.

Score guide

- 24-28: start U01 directly.
- 16-23: read the linked bridge sections for missed items first.
- Below 16: work P01 through P14 in order before U01.

Local remediation rule

Every unit lesson opens with a remediation block that re-teaches the smallest set of objects the unit uses before any other course does (for example U01 re-teaches code points and UTF-8 even though P13 covers them). The block is self-contained and assessed.

Notation and shapes

Shared symbols used across all nine units. A lesson may add local symbols, local symbols never override these.

Index conventions

- B: batch size (sequences per batch). Unit: count.
- T (or S): sequence length in tokens. Unit: count.
- d (or d_model): model width. Unit: count.
- h: number of attention heads. Unit: count.
- d_h: head dimension, $d_h = d / h$. Unit: count.
- V: vocabulary size. Unit: count.
- L: number of transformer layers. Unit: count.
- d_ff: feedforward hidden width. Unit: count.
- E: number of experts (MoE). Unit: count.
- k: top-k experts per token (MoE). Unit: count.
- N: parameter count. Unit: count.
- D: training token count. Unit: count.
- C: training compute in FLOP. Unit: FLOP.

Tensor shapes

- Token ids: x with shape (B, T), dtype int64, values in [0, V).
- Embeddings: E_tok with shape (V, d), lookup gives (B, T, d).
- Q, K, V per head: (B, h, T, d_h).
- Attention scores: (B, h, T, T) before masking.
- Attention weights: same shape, rows sum to 1 over keys.

- Logits: (B, T, V).
- KV cache per layer per token: $2 \cdot d \cdot \text{values} (K \text{ and } V), \text{ bytes} = 2 \cdot d \cdot \text{bytes_per_element}$.

Probability and loss

- p : true distribution. q (or p_{θ}): model distribution.
- $H(p, q)$: cross-entropy. Unit: nats (natural log) or bits (log base 2), the lesson states the base.
- Perplexity: $\exp(H)$ in nats. Unit: none (ratio scale).
- $KL(p \parallel q)$: Kullback-Leibler divergence. Unit: nats or bits.

Optimization

- θ : parameters. g : gradient of loss wrt θ .
- η (or lr): learning rate. Unit: $1 / (\text{loss curvature scale})$.
- t : optimizer step index. m_t, v_t : Adam first and second moments.
- β_1, β_2 : Adam decay rates. ϵ : numerical guard.
- λ : weight decay coefficient.

Hardware and systems

- FLOP: one floating-point operation. FLOP/s: rate.
- Byte counts use decimal GB (10^9) unless a lesson states GiB.
- Arithmetic intensity: FLOP per byte moved.
- Ranks: $r = 0..R-1$ for R data-parallel workers.
- Microbatch count: m , gradient accumulation steps: a .

Text and tokenization

- U+XXXX: Unicode code point in hex.
- UTF-8 bytes: sequences of 1 to 4 bytes per code point.
- Token: integer id in $[0, V)$. A token may decode to bytes, not to a full character.
- Compression rate: bytes per token, or tokens per word.

Conventions

- Shapes are written (B, T, d) in that order unless stated.
- Log means natural log unless the base is stated.
- "Not in source" marks any value not computed by a lesson script or quoted from a cited paper.

Glossary

One meaning per term. The lesson that introduces a term owns the full definition, this glossary points there.

U01 , Tokenization

- Code point: a number assigned to a character by Unicode (U+0041 = "A").
- UTF-8: a variable-length encoding of code points as 1 to 4 bytes.
- Token: an integer id from a fixed vocabulary used as model input.
- BPE (byte pair encoding): a tokenizer that starts from bytes and merges the most frequent adjacent pair, repeating to a target vocabulary size.
- Merge: one BPE step that joins a chosen pair into a new token.
- Pretokenization: splitting raw text (often by regex) before BPE merges.
- Special token: a token with control meaning (for example end of text), never produced by normal merges.
- Roundtrip: `decode(encode(text))` equals text byte-for-byte.
- Fertility: tokens produced per word, lower means better compression.

U02 , Tensor programming

- Shape contract: the promised input and output shapes of an operation.
- View: a tensor that shares storage with another tensor.
- Stride: the step in storage between consecutive elements along an axis.
- Contiguous: storage order matches logical order (stride pattern standard).
- Activation: a tensor saved during the forward pass for the backward pass.
- Arithmetic intensity: FLOPs per byte moved between memory levels.
- Roofline: a plot of attainable FLOP/s against arithmetic intensity.
- Checkpointing (activation): recompute activations in backward to save memory.

U03 , Transformer block

- Embedding: a learned vector per token id, lookup, not computation.
- Query/Key/Value (Q/K/V): linear projections of the input, attention compares queries against keys and mixes values.
- Causal mask: a mask that blocks attention to future positions.
- Head: one independent attention unit, heads concatenate their outputs.
- Residual connection: $x + \text{block}(x)$, the identity path that carries gradient.
- Pre-norm: normalization before the sub-block, post-norm: after.
- RMSNorm: normalization by root mean square, without mean centering.
- SwiGLU: a gated feedforward activation: $\text{Swish}(xW)$ elementwise-times (xV) .
- RoPE: rotary position embedding, rotates Q/K pairs by position angle.
- Weight tying: the output head reuses the input embedding matrix.

U04 , Attention alternatives and MoE

- MHA: multi-head attention, separate K/V per head.
- MQA: multi-query attention, one K/V head shared by all query heads.
- GQA: grouped-query attention, groups of query heads share K/V heads.
- MLA: multi-head latent attention, K/V compressed to a latent vector.
- Linear attention: attention rewritten with a kernel so cost grows linearly in sequence length.
- MoE: mixture of experts, a router sends each token to k of E experts.
- Router: the gating network that scores experts per token.
- Dropped token: a token an expert skips when its capacity is full.
- Load balancing: auxiliary losses that spread tokens across experts.

U05 , Optimization

- Cross-entropy loss: negative log probability of the true next token.
- Label shift: targets are inputs shifted by one position.
- Adam: adaptive optimizer with first and second moment estimates.
- AdamW: Adam with decoupled weight decay.
- Bias correction: Adam's division by $(1 - \beta^t)$ that fixes early steps.
- Warmup: learning rate rises from 0 over the first steps.
- Cosine decay: learning rate follows a cosine curve down to a floor.
- Gradient clipping: rescale the gradient when its norm exceeds a limit.
- RNG state: the random generator state saved for exact resume.

U06 , Hardware

- HBM: high-bandwidth memory on the GPU package.
- SRAM: fast on-chip memory (shared memory, registers).
- Tensor core: a matrix-multiply unit on NVIDIA GPUs.
- Occupancy: fraction of the GPU's warp slots that hold live warps.
- Coalescing: adjacent threads read adjacent memory addresses.
- Bank conflict: shared-memory accesses that serialize on the same bank.
- Launch overhead: fixed CPU cost of starting a GPU kernel.

U07 , Kernels

- Fusion: one kernel does the work of several, keeping data on chip.
- Tiling: split a large operation into SRAM-sized blocks.
- Online softmax: softmax computed in one pass with a running maximum.
- FlashAttention: IO-aware exact attention using tiling and online softmax.
- Triton: a Python-like language for writing GPU kernels.
- Grid: the set of thread blocks (Triton) or blocks (CUDA) of a kernel.
- Register pressure: too many live values per thread lowers occupancy.

U08 , Sharding

- DDP: distributed data parallel, each rank holds a full replica.
- All-reduce: every rank ends with the sum of all ranks' tensors.
- Reduce-scatter: every rank ends with one shard of the sum.
- All-gather: every rank collects all shards into the full tensor.
- ZeRO: sharding of optimizer state, gradients, and parameters.
- Microbatch: one slice of a batch processed before the optimizer step.
- Bucket: a gradient buffer chunk synchronized as one collective.

U09 , Parallelism

- Tensor parallelism: split individual layers across ranks.
- Pipeline parallelism: split layers into stages across ranks.
- Pipeline bubble: idle time while the pipeline fills and drains.
- Sequence parallelism: split the sequence dimension across ranks.
- Expert parallelism: place different experts on different ranks.

- Hybrid layout: a named combination such as DP + TP + PP.
- Straggler: a rank that lags and slows a synchronized step.

U10 , Scaling

- Scaling law: an empirical relation between compute, data, parameters, and loss, usually a power law with an irreducible floor.
- IsoFLOP curve: loss versus allocation at fixed compute budget.
- Compute-optimal: the parameter/data split that minimizes loss at a fixed compute.
- Preregistration: a written prediction with a band and a decision rule, made before the experiment runs.
- Bootstrap interval: an uncertainty band from resampling the data.

U11 , Inference

- Prefill: the first pass over the prompt, compute-bound at long lengths.
- Decode: token-by-token generation, bandwidth-bound.
- KV cache: stored keys and values that avoid recomputation.
- Continuous batching: inserting and evicting sequences at iteration boundaries instead of waiting for a whole batch.
- Speculative decoding: a small model drafts, the target verifies.
- TTFT: time to first token. TPOT: time per output token.

U12 , Evaluation

- Perplexity: e raised to the cross-entropy loss.
- Contamination: benchmark text present in training data.
- Canary: a planted unique string that detects contamination.
- Internal evaluation: steers iteration. External evaluation: judges.
- Kappa: agreement between judges corrected for chance.
- Slice: a subset of the eval that isolates one capability.

U13 , Data sourcing

- WARC: the web archive record format. CDX indexes it.
- Extraction yield: kept bytes divided by raw bytes.
- Provenance: the URL, crawl date, and license of a document.
- PII: personally identifiable information, detected and redacted.
- Version hash: a content digest that identifies a dataset build.

U14 , Dedup and mixing

- Bloom filter: a bit array with k hashes, no false negatives.
- Shingle: an n -gram used as a set element for near-dup detection.
- MinHash: a signature that estimates Jaccard similarity.
- LSH: locality-sensitive hashing that finds candidate pairs.
- Reweighting: scaling a source by target weight over source weight.
- Mixture search: choosing domain proportions for a target metric.

U15 , Midtraining and SFT

- Midtraining: continued pretraining with a new data mix.
- Loss mask: the 0/1 rule that puts loss on assistant tokens only.
- Packing: concatenating short sequences into full context windows.
- LoRA: a low-rank adapter added to frozen weights.
- Forgetting: base capability lost during adaptation, measured in nats.

U16 , Alignment and RL

- Reward model: a learned scorer of response quality.
- Bradley-Terry: the pairwise preference probability model.
- KL regularization: a penalty that keeps the policy near a reference.
- PPO: policy optimization with a clipped surrogate objective.
- GRPO: group-relative baselines without a value network.
- DPO: direct preference optimization without RL.
- Reward hacking: the proxy rising while the truth falls.

U17 , Systems

- Modality token: non-text input (image patches) as tokens.
- Interleaved stream: one token sequence mixing modalities.
- Rollout fleet: workers that generate trajectories, separate from training.
- Lineage: the hash-chained record of how an artifact was produced.
- Bottleneck: the slowest stage of a step, named by profiling.
- Amdahl bound: the speedup limit from the unoptimized fraction.

U18 , Defense

- Honesty protocol: name the source, state the inspected extent, state the unknown.
- Ablation: removing one factor to measure its effect.
- Failure diary: date, symptom, hypothesis, fix, lesson.
- Resource ledger: the compute and storage cost record.
- Replication: an independent rerun inside all seed bands.
- Oral defense ladder: define, toy, derive, implement, compare, debug, critique, design.

Cheatsheet

One page per unit, formulas first. Toy numbers from executed course scripts. HYP = hypothetical.

U01 tokenization

- UTF-8: code points $< U+0080$ one byte, longer need 2-4.
- bytes/token: English ~ 4.2 , code ~ 2.5 (toy).
- Rule: the training and inference tokenizers must match.

U02 tensor programming

- matmul: $2*m*n*k$ FLOPs, backward $\sim 2x$ forward.
- Adam memory: params + grads + 2 moments (+ master weights).
- Rule: intensity = FLOPs/byte decides bound or bandwidth.

U03 transformer

- Attention: $\text{softmax}(QK^T/\sqrt{d})V$, causal mask = -inf upper.
- SwiGLU: $x \text{ silu}(W1x) (W2x)$ gating.
- Rule: pre-norm for deep stacks.

U04 attention/MoE

- KV per token: $2*L*h_{kv}*d_h*$ bytes.
- MoE active params: top-k experts, not all.
- Rule: balance the router or experts collapse.

U05 optimization

- AdamW: decay the weights, not the gradients.
- Clip: global norm $\leq c$.
- Rule: checkpoint weights + optimizer + RNG + data position.

U06 hardware

- Roofline: $\min(\text{peak FLOPs}, \text{bandwidth} * \text{intensity})$.
- MFU: achieved / peak.
- Rule: measure first, optimize second.

U07 kernels

- SRAM » HBM speed, tile to fit.
- Online softmax: running max + running sum.
- Rule: correctness vs reference before benchmarks.

U08 sharding

- ZeRO-1: optimizer. -2: +gradients. -3: +parameters.
- Rule: overlap comms with compute (bucketing).

U09 model parallel

- TP: split layers. PP: split depth, bubble $\sim (p-1)/(m+p-1)$.
- 1F1B shrinks the bubble.
- Rule: hybrid DP/TP/PP for large runs.

U10 scaling

- $L = E + A \cdot (C/C_0)^{-a}$. $C = 6ND$.
- Toy: $a=0.34 \rightarrow 0.340$. $N=7.0e10$, $D=1.4e12$, $D/N=20$.
- Rule: preregister extrapolations with bands.

U11 inference

- KV: $2 \cdot L \cdot h_{kv} \cdot d_h \cdot \text{bytes/token}$ (toy 128 KB. 8.59 GB).
- Spec decode speedup = $E/(k \cdot c + 1)$ (toy 1.95x).
- Rule: serve to SLOs (TTFT, TPOT, p99), not to throughput.

U12 evaluation

- $\text{ppl} = e^{\text{loss}}$ (toy $e^{2.3} = 9.97$).
- $\text{SE} = \sqrt{p(1-p)/n}$ (toy 0.72 ± 0.039).
- $\text{kappa} = (\text{agree} - \text{expected}) / (1 - \text{expected})$.
- Rule: never tune on the external eval.

U13 data sourcing

- $\text{yield} = \text{kept}/\text{raw}$ (toy 28.6%).
- Provenance: URL + date + license.
- Rule: sort, seed, pin, hash, prove determinism.

U14 dedup/mixing

- Bloom FPR: $(1 - e^{-kn/m})^k$ (toy 0.0216, $k=6$).
- LSH: $P = 1 - (1 - s^r)^b$.
- Rule: dedup before filtering, dedup train vs eval.

U15 midtrain/SFT

- Mask: loss on assistant tokens only.
- LoRA: $2 \cdot d \cdot r$ per pair (toy 8.39M).
- Packing: 47.6% -> 22.0% waste (toy).
- Rule: templates byte-identical train/serve.

U16 alignment/RL

- BT: $\text{sigmoid}(r_w - r_l)$.
- PPO: $\min(rA, \text{clip}(r, 1-\text{eps}, 1+\text{eps})A)$.
- GRPO: $(r - \text{mean})/\text{std}$ per group.
- DPO: $r = \beta \cdot \log(\pi/\pi_{\text{ref}})$ (toy margin 0.139).
- Rule: watch the truth metric, not the proxy.

U17 multimodal/systems

- Image: 256 tokens (16x16 patches).
- Amdahl: $\text{new} = \text{total} - \text{share} + \text{share}/\text{speedup}$ (toy 143 ms).
- Rule: profile first, the bottleneck moves.

U18 defense

- Replication: inside all seed bands.
- Ablations: $2^n / 2^{\{n-k\}} / 1+n$.
- Rule: never infer content from a speaker's name.
- Ladder: define, toy, derive, implement, compare, debug, critique, design.

Crash course

Condensed review of the full course. Each unit: the one-line idea, the key numbers, and the most interview-tested concept. Numbers marked toy come from the course's executed numpy scripts, numbers marked HYP are hypothetical. U01-U09 by the first builder. U10-U18 in this build.

U01 , tokenization

Text becomes integers. BPE merges the most frequent pair, the pretokenization regex decides what can merge. Compression is bytes per token: ~4.2 for English web text, ~2.5 for code (toy). Mismatch between training and inference tokenizers silently changes every count.

U02 , tensor programming

Shapes are contracts. einsum states them, strides explain views. Matmul costs $2*m*n*k$ FLOPs, backward ~2x forward. Arithmetic intensity decides whether a kernel is memory-bound. Optimizer state (Adam: 2 moments) doubles the parameter memory.

U03 , transformer block

Embed, project Q/K/V, scaled softmax attention, causal mask, multi-head reshape, residual, pre-norm, SwiGLU FFN, RoPE, output head. Pre-norm stabilizes deep stacks. RMSNorm drops the recenter. RoPE rotates pairs by position, theta sets the ruler.

U04 , attention and MoE

GQA trades KV heads for memory: fewer heads, same quality, much smaller cache. MLA compresses the cache further. MoE routes each token to top-k experts, the router needs a load-balance loss or experts collapse. Active compute is per-token, not per-parameter.

U05 , optimization

AdamW decouples weight decay. Warmup then cosine (or WSD). Gradient clipping bounds the step, not the loss. Batch size and LR scale together up to the critical batch size. Checkpoint weights, optimizer moments, RNG state, and data position, or the resume is not a resume.

U06 , GPU/TPU

The roofline: $\min(\text{peak FLOPs}, \text{bandwidth} \times \text{intensity})$. Decode is bandwidth-bound (1 FLOP/byte), prefill is compute-bound at long T. MFU measures achieved vs peak. Measure before you optimize. vendor claims are branches to verify.

U07 , kernels

Tile for the memory hierarchy, keep hot data in SRAM. Online softmax makes FlashAttention possible: one pass, no materialized $N \times N$. Triton is the portable kernel language. Correctness first against a reference, benchmark claims are branches.

U08 , data sharding

Data-parallel averages gradients with all-reduce. ZeRO shards optimizer state, then gradients, then parameters. Overlap communication with backward compute via bucketing. Mixed precision needs loss scaling. Fault tolerance is a first-class design item, not an afterthought.

U09 , model parallel

Tensor-parallel splits layers (needs fast interconnect). pipeline-parallel splits depth (pays the bubble). 1F1B schedules shrink the bubble. Hybrid DP/TP/PP is the standard large-run recipe. Activation checkpointing trades recompute for memory.

U10 , scaling laws and extrapolation

Loss follows $L = E + A \cdot (C/C_0)^{-a}$ (toy: $a=0.34$, recovered 0.340, RMS 0.0031). 6ND estimates compute. The isoFLOP optimum for the toy law: $N=7.0e10$, $D=1.4e12$, $D/N=20$. Preregister every extrapolation: prediction, band, rule, date. Report bootstrap bands, not point estimates. Extrapolation fails on form change, noise regime, and range.

U11 , inference and serving

Decode is bandwidth-bound: each token re-reads the weights. KV cache: $2 \cdot L \cdot h_{kv} \cdot d_h \cdot \text{bytes/token}$ (toy: 128 KB/token, batch $8 \times 8192 = 8.59$ GB). Continuous batching beats static batching. Speculative decoding: $\text{speedup} = E/(k \cdot c + 1)$ (toy: 1.95x). Serve with SLOs: TTFT, TPOT, p99. The cost/latency/quality triangle rules every decision.

U12 , evaluation and validity

Perplexity = $e^{\{\text{loss}\}}$ (toy: $e^{2.3} = 9.97$). Always report CIs: $SE = \sqrt{p(1-p)/n}$ (toy: 0.72 ± 0.039 at $n=500$). Dedup train against eval: n-gram overlap + canaries (toy: 100% overlap found, recall 1.00). Internal evals steer, external evals judge. never tune on the external. Judge agreement needs kappa, not raw percent. Metrics must carry cost.

U13 , data sourcing

WARC stores the crawl. CDX indexes it. Extraction yield = kept / raw bytes (toy: $1.9/6.6$ GB = 28.6%). Hazards: PDFs, code files, book chapters. Provenance: URL + crawl date + license. Language ID: the non-English-called-English rate is the metric (toy: 2.8%). PII: detect, redact, never train on it. Version with content hashes (toy id 613575a7e17d78eb), sort, seed, pin, hash to prove determinism.

U14 , filtering, dedup, mixing

Dedup first, then filter: scores computed once per unique doc. Bloom: $(1 - e^{-kn/m})^k$, no false negatives (toy: $k=6$, FPR 0.0216 at 8 bits/item). Near-dup: shingles $\rightarrow$ MinHash $\rightarrow$ LSH bands. $P(\text{match}) = 1 - (1 - s^r)^b$ (toy: Jaccard 0.113 vs 0.125). Train-test dedup at n-gram level (toy: 13.3% overlap). Mixing: reweight by target/source ratio, interaction terms move the optimum (toy argmin shifted to the boundary, then interior). Repetition trades recall for memorization (toy logistic).

U15 , midtraining and SFT

Stages: pretrain (LM), midtrain (continued LM, new mix), post-train (behavior). RoPE theta 50x stretches the ruler but needs long documents. Train/serve templates must match byte-exactly. SFT loss on assistant tokens only. Packing cuts padding waste 47.6% $\rightarrow$ 22.0% (toy, 344 bins) with boundary masks. LoRA: $2 \cdot d \cdot r$ per pair (toy: 8.39M at $r=16$). Measure forgetting: mean +0.15 nats over 6 checkpoints (toy). Full checkpoints carry weights, optimizer, RNG, data position.

U16 , alignment and RL

BT: $P(\text{pref}) = \text{sigmoid}(r_w - r_l)$. KL keeps the policy near the reference. PPO: $\min(r \cdot A, \text{clip}(r) \cdot A)$ (toy terms [0.7, 0.9, -1.0, 1.2, 1.2]). GRPO: group mean/std as baseline (toy: [1,-1,1,-1] vs [0,0,0,0]). DPO: $r = \beta \cdot \log(\pi/\pi_{\text{ref}})$ inside the BT loss (toy: margin 0.139, loss 0.626). Verifiable rewards beat learned rewards where checkable. Reward hacking: proxy up, truth down. the held-out truth metric reveals it. Collapse = all-equal advantages = zero signal.

U17 , multimodality and systems

Images become tokens: 16x16 patches $\rightarrow$ 256 tokens each. Budget tokens against the context (toy: 8 images + 512 text = 2560, 80% image). Interleaved streams mix modalities with boundary tokens. Interfaces: tok/s/GPU, GB/s, bytes/step. Separate rollout and training fleets, async with bounded queues (toy: 83.3% util, 1200/hr backlog at 120 vs 100/min). Lineage: $\text{id}_n = \text{sha256}(\text{id}_{n-1} + \text{stage}_n)$. Amdahl: new total = total - share + share/speedup (toy: 143 ms, backward 38.5%, 2x $\rightarrow$ 19.2%).

U18 , guests and defense

Guest content is unknown: no artifacts inspected, no inference from speaker names. The honesty protocol: name the source, state the inspected extent, state the unknown. Replication passes when an independent rerun lands inside all seed bands. Ablations: full 2^n , fractional $2^{\{n-k\}}$, OAT $1+n$ (toy: 16/8/5 for 4 factors). Keep a failure diary: date, symptom, hypothesis, fix, lesson. Report cost honestly (toy: \$1,200 for 480 GPU-hr. 25% failure share). The oral defense ladder: define, toy, derive, implement, compare, debug, critique, design.

Capstone A , research replication (honest failure)

Replicate the U10 power-law fit on a new synthetic law, then test a preregistered extension at 4x noise. Replication: true $a=0.41$, fitted 0.412, PASS. Extension: observed 0.405 against the preregistered band $[0.30, 0.38]$, FAILED. The failure is experimenter error, not noise: the band was anchored to the old law ($a=0.34$), not the new one ($a=0.41$). The post-hoc diagnosis is labeled post-hoc. The preregistered test stands as failed. Lesson: preregister the band against the law under test, and report the miss exactly as registered.

Capstone B , applied serving plan (all inputs HYPOTHETICAL)

Fictional SaaS support chat on a 7B bf16 model. SLOs (HYP): p99 TTFT < 2000 ms, p99 TPOT < 50 ms at 40 peak RPS (HYP). Sizing (computed from HYP inputs): 1.7 rps/GPU, 32 GPUs at 30% headroom, 14.9 GB of 80 GB per GPU, \$57,600/month (HYP). Rollout: 1% canary 24h, then 10%, 50%, 100% on green gates. Rollback on +0.5pt error rate or +20% p99 latency. Chosen for v1: full-precision 7B at 32 GPUs, quantize only after U12 evals. Nothing here is measured.

The arc in one paragraph

Tokens become tensors (U01-U02), tensors become a transformer (U03-U04), trained by Adam on sharded hardware (U05-U09), following scaling laws (U10), served with caches and batching (U11), measured with valid evals (U12), fed by sourced, deduped, mixed data (U13-U14), adapted by midtraining and SFT (U15), aligned by RL (U16), extended to images and run as a system (U17), and defended as honest research (U18).